

UTSJOKI NUORGAM, Finland

WMO: 028250

Lat: 70.0819N

Lon: 27.8964E

Elev: 72

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-19.8	-31.8	0.9	-25.0	-25.6	1.3	-19.8	24.9	23.0	20.8	23.3	1.9	200	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	73.9	61.3	69.1	59.3	65.1	57.2	63.1	71.3	60.3	67.1	58.0	63.8	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	75.9	66.4	57.1	69.7	63.8	54.9	64.2	61.2	28.6	72.5	26.7	67.0	25.1	64.5	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	14.8	12.6	DB	-30.3	78.9	7.0	4.9	-35.3	82.5	-39.4	85.3	-43.4	88.1	-48.5	91.7	(4)
(5)			WB	-30.5	65.3	6.6	3.8	-35.3	68.0	-39.1	70.3	-42.8	72.4	-47.6	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	9.8	11.6	17.7	28.5	40.3	47.9	54.4	50.7	44.8	32.3	21.8	15.9
	DBStd	18.13	13.88	14.77	11.12	8.84	6.39	6.42	6.42	5.89	5.70	8.05	10.71	12.23
	HDD50	7110	1248	1076	1000	646	311	116	24	62	174	550	846	1057
	HDD65	12266	1713	1496	1465	1096	765	514	335	443	606	1015	1296	1522
	CDD50	327	0	0	0	0	11	53	161	85	17	0	0	0
	CDD65	9	0	0	0	0	0	1	7	1	0	0	0	0
	CDH74	153	0	0	0	0	10	12	114	17	0	0	0	0
	CDH80	30	0	0	0	0	3	0	23	4	0	0	0	0
Wind	WSAvg	5.3	5.1	5.6	5.9	5.6	6.2	6.2	5.4	4.7	4.8	4.5	4.7	5.1
Precipitation	PrecAvg	17.6	1.2	1.1	1.1	0.8	1.0	2.2	2.1	1.9	1.4	1.9	1.3	1.4
	PrecMax	24.1	2.2	2.1	1.8	1.5	2.0	4.1	5.5	3.1	3.1	4.2	2.2	2.4
	PrecMin	13.0	0.3	0.2	0.4	0.1	0.4	0.3	0.6	0.9	0.3	0.3	0.4	0.7
	PrecStd	2.6	0.6	0.5	0.4	0.3	0.5	1.0	1.4	0.7	0.7	1.0	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	39.0	40.0	49.0	72.0	75.3	83.5	75.6	65.0	52.1	41.2	40.5
		MCWB	35.6	36.1	36.1	42.9	55.1	61.3	66.0	62.1	57.5	48.0	39.6	37.2
	2%	DB	35.2	36.3	37.5	43.7	62.5	68.8	76.9	69.7	61.3	48.8	38.7	36.3
		MCWB	32.9	34.0	34.5	38.6	50.4	57.4	63.3	60.5	54.9	45.7	36.9	34.3
	5%	DB	30.9	33.5	35.2	41.3	56.4	63.9	71.4	65.8	58.0	45.7	36.4	34.5
		MCWB	29.2	31.3	32.5	36.9	47.4	55.0	60.9	57.9	52.8	43.7	34.9	32.7
	10%	DB	27.8	30.5	33.1	39.5	51.5	59.5	66.6	61.7	54.9	43.1	34.7	32.4
		MCWB	26.4	28.8	30.7	35.8	44.2	52.2	59.0	56.0	50.9	41.5	33.3	30.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	36.4	36.6	43.0	56.8	62.6	67.3	64.7	58.7	48.3	40.2	37.4
		MCDB	38.1	38.4	39.5	49.0	69.9	71.6	78.8	71.5	63.5	51.6	41.2	40.3
	2%	WB	33.1	34.1	34.7	38.8	50.7	59.4	64.7	61.2	56.0	46.0	37.1	34.7
		MCDB	35.0	36.1	37.3	42.6	60.4	66.2	74.5	67.9	60.6	48.6	38.4	36.1
	5%	WB	29.3	31.5	32.8	37.5	47.8	55.9	62.0	58.8	53.3	43.8	35.0	32.8
		MCDB	30.7	33.3	35.0	40.7	55.7	63.3	69.8	64.1	56.9	45.5	36.3	34.4
	10%	WB	26.4	28.8	30.8	36.2	45.1	52.8	59.2	56.6	51.3	41.5	33.3	30.7
		MCDB	27.7	30.3	33.0	39.0	50.6	58.7	65.6	61.0	54.7	43.1	34.5	32.2
Mean Daily Temperature Range	5% DB	MDBR	17.3	15.8	18.2	15.9	14.3	13.9	15.4	15.7	14.7	10.7	12.8	15.8
		MCDBR	15.8	11.8	11.9	13.1	25.1	21.9	25.5	23.1	20.1	10.9	10.1	12.5
		MCWBR	14.8	10.4	10.3	9.1	14.4	12.1	13.2	14.1	13.8	8.9	9.4	11.8
	5% WB	MCDBR	15.8	12.3	11.6	12.1	24.1	20.3	23.2	20.2	16.8	10.2	10.3	12.4
		MCWBR	14.9	11.1	10.2	8.6	14.0	11.9	13.1	13.0	12.0	8.5	9.7	11.8
Clear-Sky Solar Irradiance	taub	N/A	0.252	0.302	0.329	0.351	0.347	0.361	0.343	0.311	0.263	N/A	N/A	
	taud	N/A	2.018	2.088	2.146	2.298	2.444	2.436	2.466	2.509	2.181	N/A	N/A	
	Ebn at Noon	0	140	215	251	262	268	259	250	226	138	0	0	
	Edh at Noon	0	13	26	33	33	30	29	25	18	11	0	0	
All-Sky Solar Radiation	RadAvg	0	95	486	1059	1433	1515	1380	941	489	141	0	0	
	RadStd	0	10	34	72	143	124	143	90	44	16	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page